QUIZ

NumPy & Pandas

Python Data Science Series

STUDENT INFORMATION

FULL NAME

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CLASS / SECTION

7

ROLL NUMBER

24scse2240036

Date: _____

Total Marks: ____ / 30

SECTION 1

NumPy Arrays & Operations

Questions 1–5 · Broadcasting, Indexing & Numerical Computing

5 Questions × 1 Mark = 5 Marks

1

ARRAY OPERATIONS

[1 Mark]

What does the following NumPy operation return?
np.array([1, 2, 3]) * 3

A [3, 6, 9]

B [1, 2, 3, 1, 2, 3, 1, 2, 3]

C [3, 3, 3]

D Error – cannot multiply array by scalar

Answer: A

2

BROADCASTING

[1 Mark]

Which of the following NumPy array shapes are broadcast-compatible?

A	(3, 4) and (5, 4)
B	(3,) and (3, 3)
C	(4, 1) and (1, 4)
D	Both B and C

Answer: ____D___

3

INDEXING & SLICING

[1 Mark]

Given arr = np.array([[1,2,3],[4,5,6],[7,8,9]]), what does arr[1:, ::2] return?

A [[4, 6], [7, 9]]

B [[4, 5, 6], [7, 8, 9]]

C [[4, 6], [7, 9]]

D [[5, 6], [8, 9]]

Answer: C

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NUMERICAL COMPUTATIONS

[1 Mark]

Which NumPy function computes the dot product of two 1-D arrays?

A np.multiply()

B np.dot()

C np.cross()

D np.sum()

Answer: _____

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ARRAY OPERATIONS

[1 Mark]

What is the output of np.arange(2, 10, 3)?

A	[2, 5, 8, 11]
B	[2, 5, 8]
C	[3, 6, 9]
D	[2, 4, 6, 8]

Answer: ____B___

SECTION 2

Pandas — Series & DataFrames

Questions 6–12 · Data Loading, Filtering, Sorting & Wrangling

7 Questions × 1 Mark = 7 Marks + 3 Short Answer × 2 Marks = 6 Marks

Which Pandas method is used to read a CSV file into a DataFrame?

A	pd.read_table()
B	pd.read_csv()
C	pd.load_csv()
D	pd.import_csv()

Answer: ____B___

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DATA LOADING

[1 Mark]

Which parameter in pd.read_excel() specifies which sheet to load?

A tab_name

B sheet_index

C sheet_name

D worksheet

Answer: ___C____

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FILTERING

[1 Mark]

Given df with a column 'Age', which expression filters rows where Age > 25?

A df[df.Age > 25]

B df.filter(Age > 25)

C df.where('Age > 25')

D df.query['Age > 25']

Answer: _____A__

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SORTING & GROUPING

[1 Mark]

Which method sorts a DataFrame by the 'Score' column in descending order?

A df.sort('Score', desc=True)

B df.order_by('Score', ascending=False)

C df.sort_values('Score', ascending=False)

D df.arrange('Score', reverse=True)

Answer: __C_____

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GROUPING

[1 Mark]

What does df.groupby('Dept')['Salary'].mean() return?

A A single mean salary value

B Mean salary grouped by department

C Mean salary of the 'Dept' column

D Error – cannot group then mean

Answer: ____B___

Which method combines two DataFrames horizontally (column-wise)?

A pd.concat([df1, df2], axis=0)

B pd.concat([df1, df2], axis=1)

C df1.append(df2)

D pd.merge(df1, df2, how='outer')

Answer: ___B____

What does df.dropna() do to a DataFrame?

A Drops all columns with NaN values

B Fills NaN values with 0

C Drops all rows containing NaN values

D Replaces NaN with the column mean

Answer: ___C____

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PANDAS – GROUPBY

[2 Marks]

SHORT ANSWER

Explain the split-apply-combine strategy used by groupby() in Pandas. Provide a real-world example showing how groupby() with an aggregation function is used.

Answer: Split will group the data in the given column
Apply will apply operations on the grouped data, like .max(), .mean()
Combine will combine both split results with apply.

df.groupby[“Department”](“Salary”).mean()
Here, we will group the data by department and calculate the mean salary for each department.

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NUMPY – BROADCASTING

[2 Marks]

SHORT ANSWER

What is broadcasting in NumPy? State the broadcasting rules and give an example showing how arrays of shapes (3, 1) and (1, 4) would interact.

Answer: Broadcasting is applying the changes all over the array.
Applying any change by simply adding, subtracting, multiplication to the array and it will compatible if their dimensions match or one of them is 1.2. Dimension extension: If arrays have different numbers of dimensions, the smaller array's shape is padded with 1s on the left.3. Size matching: In each dimension, the sizes must either be the same or one must be 1 (the array with size 1 is stretched to match the other).

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DATA WRANGLING

[2 Marks]

SHORT ANSWER

Describe any TWO common data-wrangling tasks in Pandas. For each task, name the method used and write a one-line code example.

Answer: 1. Handling missing values: The fillna() method is used to fill or replace missing (NaN) values in a DataFrame. 2. Removing duplicates: The drop_duplicates() method is used to remove duplicate rows from a DataFrame.Code .

END OF QUIZ

Please review all answers before submitting.

Section	Topic	Questions	Marks
Section 1	NumPy Arrays & Operations	Q1–Q5	5 Marks
Section 2	Pandas MCQ	Q6–Q12	7 Marks
Section 2	Short Answer	Q13–Q15	6 Marks
	TOTAL		**30 Marks**